Membership Inference Attacks on Discrete Diffusion Language Models: Mid-Term Progress Report

Shailesh K
me22b192@smail.iitm.ac.in
Indian Institute of Technology Madras

Abstract

Discrete Diffusion Language Models (DLMs) have emerged as a promising non-autoregressive paradigm for text generation, yet their privacy properties under fine-tuning remain largely unstudied. We investigate membership inference attacks (MIAs) against a fine-tuned Qwen3-0.6B masked diffusion model using the MIMIR GitHub benchmark. Two experiments are reported: an exploratory run that uncovered a critical mask-token-id incompatibility in the SAMA attack codebase, and a main experiment that fixed this issue and scaled the setup to 2 000 balanced samples. Our white-box feature classifier—a 112-dimensional bundle drawn from the model's noise-level loss profile, predictive entropy, denoising consistency, and cross-model hidden-state similarity—achieves AUC = 0.856, surpassing both SAMA (AUC = 0.806) and scalar-score baselines (AUC $\leq$ 0.715). At the practically relevant 0.1% FPR operating point, XGBoost recovers 17.1% of true members—roughly $40\times$ more than SAMA at the same false-positive budget.

1 Introduction

Language models memorise portions of their training data [Carlini et al., 2021], and an adversary can exploit this to infer whether a specific text was used during training—a *membership inference attack* (MIA) [Shokri et al., 2017]. While MIAs are well studied for autoregressive (AR) models [Carlini et al., 2022, Mattern et al., 2023], the growing class of *discrete diffusion language models* (DLMs) [Austin et al., 2021, Sahoo et al., 2024, Lou et al., 2024, Nie et al., 2025] has received almost no attention from a privacy standpoint.

DLMs generate text by progressively *denoising* a masked token sequence rather than predicting tokens left-to-right. Their internal signals differ structurally from AR models: the model must reconstruct tokens across the entire noise spectrum, producing richer per-sample trajectories that may encode membership information in ways that scalar-score baselines miss. Concurrently, Wei et al. [2023] establish that per-instance differential privacy leakage in discrete diffusion models is non-uniform across samples and concentrates in the low-noise regime—motivating a closer look at the noise-level loss trajectory as a privacy signal.

Contributions. (i) We identify and fix a mask-token-id mismatch that caused the SAMA attack [Chen et al., 2026] to fail silently against Qwen3-based DLMs. (ii) We propose a 112-dimensional white-box feature attack and show it substantially outperforms all scalar-score baselines, especially at low FPR. (iii) We outline a concrete future work agenda connecting DLM fine-tuning to the pDP framework of Wei et al. [2023].

2 Background

Absorbing-state discrete diffusion (D3PM, Austin et al. 2021). The forward process corrupts x_0 via a column-stochastic transition matrix Q_t, placing probability mass on a special $[\text{MASK}]$ token:

$$q(x_t \mid x_0) = \mathrm{Cat}(x_t;\ \bar{Q}_t\,\mathbf{e}_{x_0}), \quad (Q_t)_{ij} = \begin{cases} 1-\beta_t & i=j\neq[\text{MASK}], \\ \beta_t & j=[\text{MASK}], \\ 1 & i=j=[\text{MASK}]. \end{cases} \tag{1}$$

The variational bound sums KL terms over T steps: $\mathcal{L}_{\mathrm{D3PM}} = \sum_{t=2}^{T}\mathbb{E}[\mathrm{KL}(q(x_{t-1}|x_t,x_0)\,\|\,p_\theta(x_{t-1}|x_t))] - \mathbb{E}[\log p_\theta(x_0|x_1)]$.

MDLM (Sahoo et al. 2024). Switching to continuous time with the linear schedule $\alpha(t)=1-t$ collapses the ELBO to a single cross-entropy on masked positions:

$$\mathcal{L}_{\mathrm{MDLM}} = -\mathop{\mathbb{E}}_{t\sim\mathcal{U}[0,1],\, x_t\sim q(\cdot|x_0)}\big[\log p_\theta(x_0 \mid x_t)\big]. \tag{2}$$

This is the objective used to fine-tune and to estimate NLL throughout.

SEDD (Lou et al. 2024). Frames absorbing diffusion via a CTMC with rate matrix $\mathbf{Q}(t)$; learns the *concrete score* $s_\theta(\mathbf{x},t)\approx\nabla_{\mathbf{x}}\log p(\mathbf{x},t)$ by minimising a score-entropy loss, providing the theoretical underpinning for continuous-time discrete denoising.

LLaDA (Nie et al. 2025). Scales masked diffusion to 8 B parameters, demonstrating that DLMs can match AR perplexity. A key empirical finding—that predictions become more confident (lower entropy) at smaller t—motivates our per-timestep feature extraction.

MIA baselines for DLMs. For a fine-tuned model θ and base reference θ_0, three scalar scores are computed via Monte Carlo NLL estimation ($K{=}3$ samples, $t_k\sim\mathcal{U}[0,1]$):

$$s_{\mathrm{Loss}}(x) = -\widehat{\mathrm{NLL}}_\theta(x), \quad s_{\mathrm{Zlib}}(x) = -\widehat{\mathrm{NLL}}_\theta(x)\big/H_{\mathrm{zlib}}(x), \quad s_{\mathrm{Ratio}}(x) = -\widehat{\mathrm{NLL}}_\theta(x)\big/\widehat{\mathrm{NLL}}_{\theta_0}(x). \tag{3}$$

SAMA (Chen et al. 2026). Averages over M random sub-sequences of x to reduce variance:

$$\phi(x) = \frac{1}{M}\sum_{m=1}^{M}\big[\mathrm{NLL}_{\theta_0}(x^{(m)}) - \mathrm{NLL}_\theta(x^{(m)})\big]. \tag{4}$$

A higher ϕ indicates stronger membership—the fine-tuned model assigns lower NLL to member sub-sequences relative to the base model.

Per-instance DP (Wei et al. 2023). Wei et al. [2023] show that in discrete denoising diffusion, privacy leakage is not uniform: it varies per training sample based on how isolated that sample is relative to the training distribution, and concentrates in the low-noise (small-t) regime. This motivates our feature extraction at multiple noise levels and the future work directions in Section 5.

3 Methodology

Model and data. We use `dllm-hub/Qwen3-0.6B-diffusion-mdlm-v0.1` [Qwen Team, 2025], a 0.6 B-parameter DLM pre-trained on OpenWebText with the MDLM objective. Its special mask token has id 151 669 (distinct from LLaDA's 126 336). Membership labels come from the MIMIR GitHub benchmark [Duan et al., 2024].

Fine-tuning. We implement the MDLM training loop directly (Eq. 2): per mini-batch, sample $t_i\sim\mathcal{U}[\epsilon,1]$, mask each non-padding token with probability t_i, and minimise CE on masked positions. AdamW, $\eta{=}10^{-4}$, weight decay 0.01, gradient clip 1.0.

White-box feature vector. For each sample x_0 we evaluate the fine-tuned model at $T = 10$ uniformly spaced noise levels $\{t_k\}$ and record four signals per level: (1) ELBO loss $\mathcal{L}(t_k)$, (2) predictive entropy $H_{\mathrm{pred}}(t_k)$ of the mean token distribution over masked positions, (3) denoising consistency $c(t_k)$ (probability that argmax predictions agree across two independent masking draws), (4) attention entropy. These $4 \times 10 = 40$ features, one scalar aggregation (area under the loss profile), and the cross-model cosine similarity $\rho = \cos(\bar{h}_\theta(x), \bar{h}_{\theta_0}(x))$ between mean layer-wise hidden norms of the fine-tuned and base models give $\mathbf{f}(x) \in \mathbb{R}^{112}$.

Classifiers. XGBoost (200 estimators, depth 4, lr 0.05) and an MLP (256-128-64, early stopping) are trained on the feature matrix with **5-fold stratified CV**, producing out-of-fold probability scores to prevent information leakage. All reported metrics use $B = 1\,000$ bootstrap confidence intervals.

4 Experiments and Results

4.1 Experiment 1: Exploratory Pipeline Validation

Purpose. Before scaling up, we ran a small, aggressive experiment to validate the end-to-end pipeline and surface any integration issues. 300 member/non-member pairs from MIMIR GitHub split `ngram_13_0.2` were tokenised to 128 tokens each. The model was fine-tuned for 15 epochs with $3\times$ data repetition (45 effective passes), deliberately inducing strong memorisation.

Outcome. The training loss fell from 3.856 to 1.585 nats (59% drop). Memorisation was confirmed (ELBO gap = 4.625 nats for members). SAMA, however, returned AUC = 0.369—below random chance.

Root cause: mask-token-id mismatch. SAMA's `get_model_nll_params()` defaults to LLaDA's mask id 126 336 for any unrecognised model type (such as Qwen3's `a2d-qwen3`). With the wrong mask id, *both* the fine-tuned and reference model NLLs are computed under incorrect masking, making the score difference in Eq. 4 incoherent. Additionally, SAMA's post-initialisation `ref_mask_id` was set by an internal call to `get_model_nll_params(ref_model)` and was not exposed in the public config interface—so passing `model_mask_id=151669` in the config dict fixed the target model but not the reference. Over-memorisation likely amplified this by collapsing the score distributions, but the primary driver was the wrong mask id.

4.2 Experiment 2: Main Experiment with Fixes

Setup. We applied three fixes from Experiment 1: (a) a post-initialisation patch that forces `attack_obj.ref_mask_id` $\leftarrow$ 151 669; (b) switched to MIMIR split `ngram_13_0.8` (80% n-gram overlap threshold—stricter member/non-member separation); (c) scaled to 1 000 samples per class (256 tokens each) with 5 fine-tuning epochs ($1\times$ repetition), targeting a moderate memorisation regime.

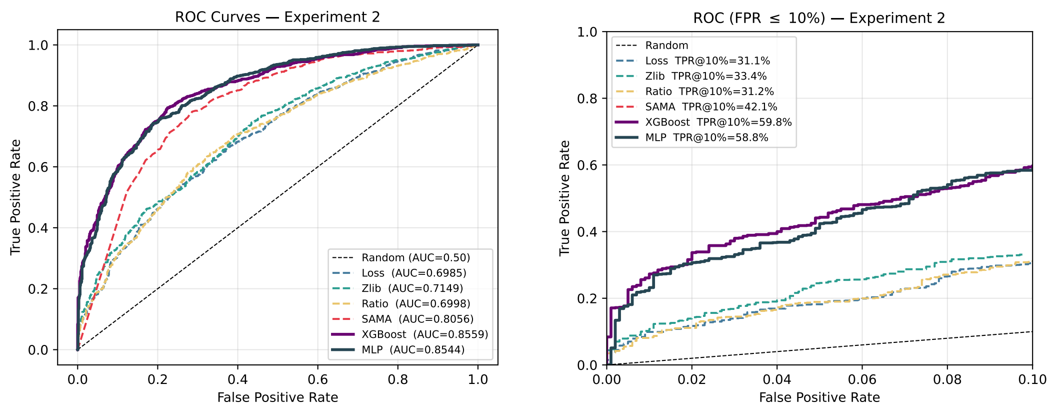
Memorisation check. ELBO gap = 2.766 nats for members vs. 1.797 nats for non-members—clear differential memorisation without the collapse seen in Experiment 1.

Benchmark results. Table 1 reports AUC and TPR at three strict FPR thresholds for all six methods on the 2 000-sample balanced test set.

Table 1: Membership inference results on MIMIR GitHub (`ngram_13_0.8`), $N = 2\,000$. Bootstrap means; **bold** = best per column.

	Method	AUC	TPR@0.1%	TPR@1%	TPR@10%
Baselines	Loss	0.6985	4.0%	9.9%	31.1%
	Zlib	0.7149	4.7%	10.8%	33.4%
	Ratio	0.6998	3.7%	8.3%	31.2%
SAMA	SAMA [Chen et al., 2026]	0.8056	0.4%	4.2%	42.1%
Ours	XGBoost	**0.8559**	**17.1%**	**27.3%**	**59.8%**
	MLP	0.8544	5.1%	23.2%	58.8%

Figure 1 shows full and low-FPR ROC curves. The classifiers (solid) dominate across the entire FPR range. SAMA achieves the best baseline AUC but near-zero TPR at FPR $\leq 1\%$; XGBoost achieves 17.1% TPR at 0.1% FPR—**40$\times$ more true positives** at the same false-positive budget. Figure 2 confirms this advantage is consistent across all three FPR thresholds.



(a) Full ROC (FPR $\in [0, 1]$) (b) Zoomed ROC (FPR $\leq 10\%$)

Figure 1: ROC curves for Experiment 2. Solid lines: our classifiers. Dashed: baselines. The zoomed panel shows the classifiers dominate in the practically relevant low-FPR regime.

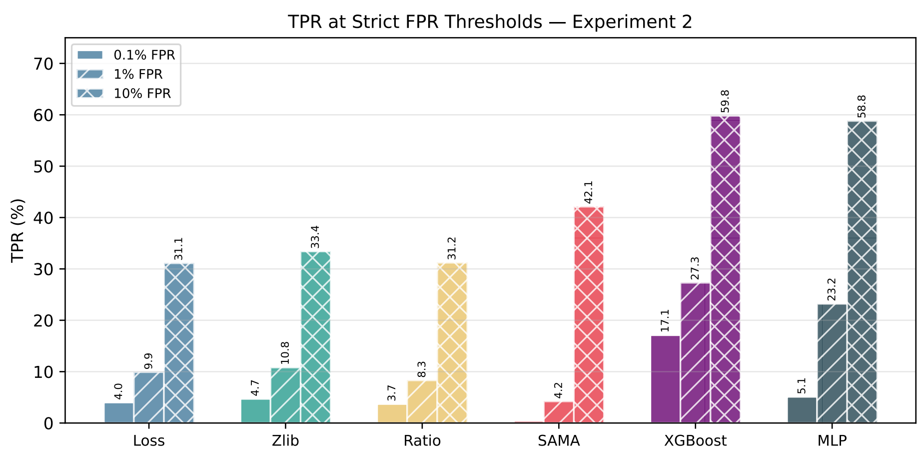


Figure 2: TPR (%) at 0.1%, 1%, and 10% FPR for all methods (Experiment 2). XGBoost and MLP (rightmost bars) achieve substantially higher TPR at every threshold, with the advantage largest at 0.1% FPR.

5 Future Work

Complete SAMA-paper benchmark. Experiment 2 validates our pipeline on a single MIMIR subset and model. We will replicate the full evaluation from Chen et al. [2026]—all nine MIMIR domain subsets (github, wikipedia, arxiv, pile-cc, dm_mathematics, hackernews, pubmed, europarl, enron) plus the additional datasets used in the SAMA paper—and extend to larger DLMs (e.g. LLaDA-8B [Nie et al., 2025]).

Per-instance privacy auditing (Wei et al.). Wei et al. [2023] establish that per-instance DP leakage in discrete diffusion correlates with how *isolated* each training sample is relative to the distribution. An open question is whether empirical MIA success rates against fine-tuned DLMs exhibit the same per-sample structure—and whether attack scores (e.g. our 112-dim features) can serve as practical proxies for the theoretical ε_i-DP leakage quantities they characterise.

Masking-ratio trajectory as a privacy signal. Wei et al. [2023] show that leakage concentrates in the low-noise regime. In masked diffusion, the per-sample loss curve across masking ratios (already partially captured by our feature vector) may therefore encode richer privacy information than a scalar NLL. Investigating the *shape* of this trajectory—and whether it improves membership classification beyond our current features—is a natural next step.

Theoretical pDP for absorbing-state DLMs. The analysis of Wei et al. [2023] is developed for categorical-noise discrete diffusion; masked diffusion uses a structurally different absorbing-state forward process (Eq. 1). Whether the pDP bounds and qualitative leakage insights carry over to the absorbing-state setting is an open theoretical problem worth pursuing.

Masking-structure-aware private fine-tuning. Dockhorn et al. [2023] show that the iterative structure of continuous diffusion enables a *noise-multiplicity* technique that improves DP-SGD privacy-utility trade-offs. The discrete masking structure of MDLM may admit an analogous exploitation: multiple independent masking realisations of the same sequence could be used as multiple gradient contributions within a single DP-SGD step, potentially improving over standard DP-LoRA baselines.

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A Additional Figures

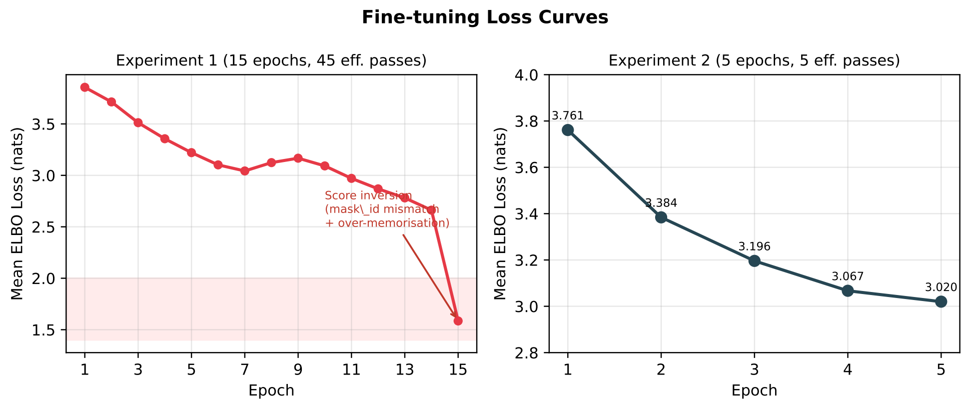

Figure 3: Fine-tuning loss curves. **Left**: Experiment 1 (15 epochs, 45 effective passes). The annotation marks the AUC inversion caused by the mask-token-id mismatch (amplified by over-memorisation). **Right**: Experiment 2 (5 epochs); the loss plateaus above 3.0 nats, consistent with moderate—rather than pathological—memorisation.

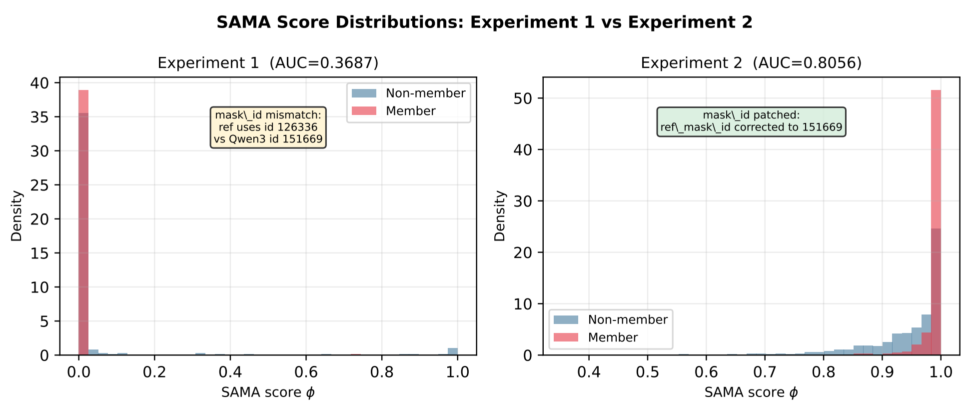

Figure 4: SAMA score distributions for members (red) and non-members (blue). **Left**: Experiment 1 — nearly identical distributions with slight inversion (AUC = 0.369), caused by the mask-token-id mismatch. **Right**: Experiment 2 — clear separation (AUC = 0.806) after patching `ref_mask_id` to 151 669.

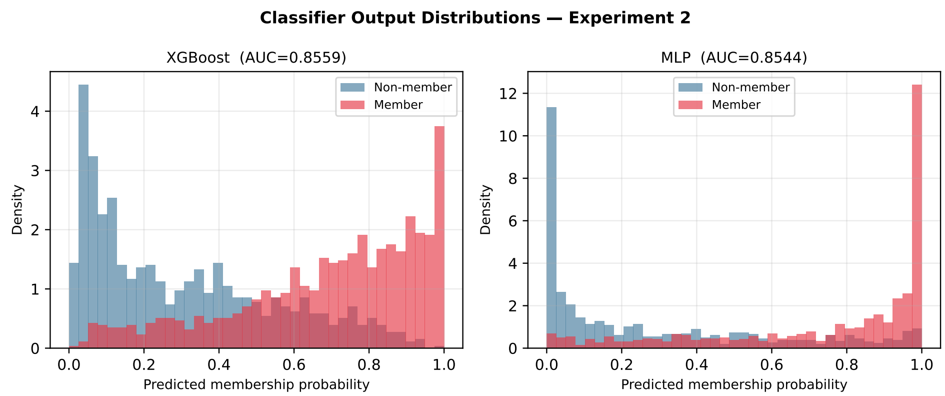

Figure 5: Predicted membership probability distributions for XGBoost (left) and MLP (right) in Experiment 2. Both classifiers separate members (red) from non-members (blue); XGBoost shows a sharper peak near 1.0 for members, explaining its higher TPR at strict FPR thresholds.

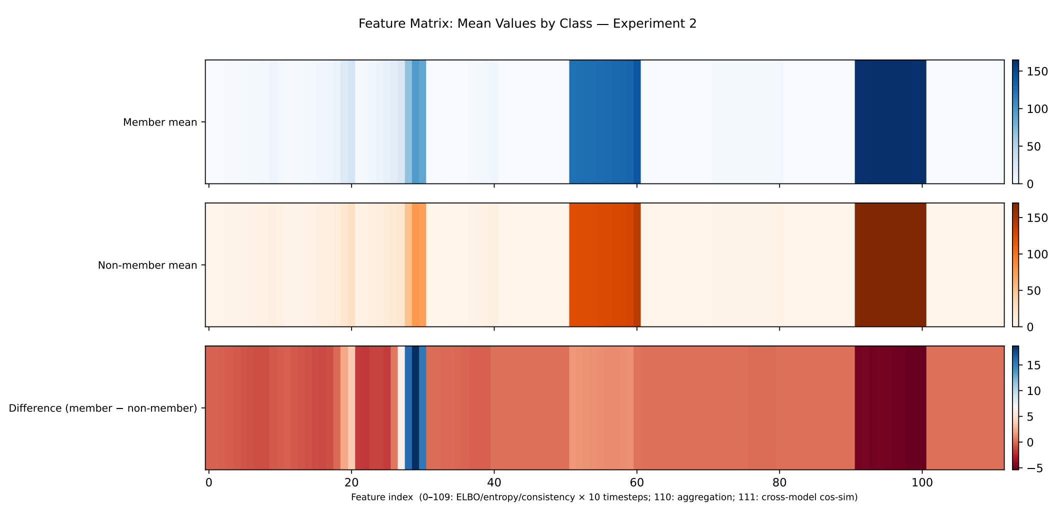

Figure 6: Mean feature values per class and their signed difference. Indices 0–39: ELBO loss at 10 timesteps; 40–79: predictive entropy; 80–109: denoising consistency; 110: area-under-loss-curve; 111: cross-model cosine similarity. The strongest discriminative signal concentrates at low noise levels (indices 0–9 and 80–89), consistent with Wei et al. [2023]'s finding that leakage concentrates in the low-noise regime.

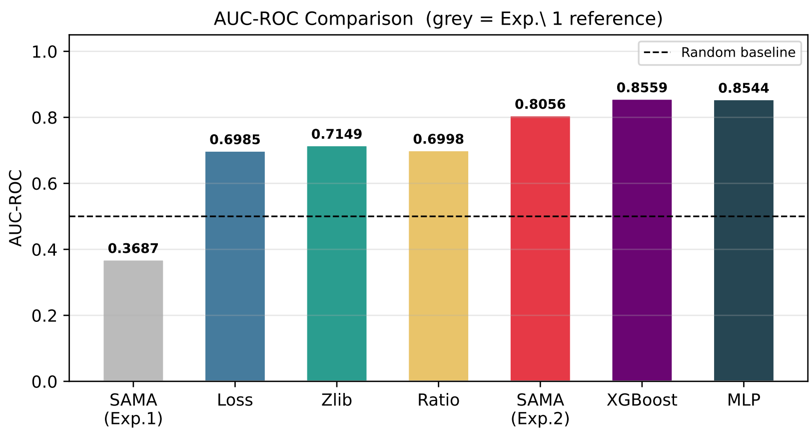

Figure 7: AUC-ROC for all methods. The grey bar (Experiment 1 SAMA = 0.369) is shown as a historical reference, illustrating the mask-token-id failure. Experiment 2 resolves this: SAMA reaches 0.806 and our classifiers push further to 0.856.